

Robert M Bowers

Research Scientist: Genomics
US Department of Energy (DOE)
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QUALIFICATION SUMMARY

Microbial Genomics Scientist with 15+ years of experience in academia, biotech and government. Proven track record designing, optimizing and validating high-throughput sequencing protocols and bioinformatic pipelines for isolate, single-cell and metagenomic datasets. Skilled in applying robust statistical methods and rigorous quality-control standards to interpret high-dimensional NGS and cell-free DNA data. Led end-to-end projects—from assay development for airborne and blood-borne pathogens to containerized R/Python workflows on HPC and cloud—and translated findings into 40+ peer-reviewed publications. Adept at building interactive R-Shiny dashboards for real-time data exploration, mentoring junior scientists, and driving cross-functional collaborations that advance both basic research and diagnostic applications.

Key Achievements

- **Publication record:** 40+ articles in top journals (ISME J, Nat Biotechnol, Microbiome).
- **Protocol & pipeline design:** Developed and validated high-throughput sequencing workflows and bioinformatic pipelines for isolate genomes, single-cell genomics, and metagenomes.
- **Assay development:** Optimized high-sensitivity molecular assays for airborne viruses and bacterial pathogens.
- **Data analysis & QC:** Applied established statistical frameworks and institutional QC standards to interpret complex NGS and cell-free DNA datasets.
- **Computational infrastructure:** Containerized R/Python pipelines for HPC and cloud; built interactive R-Shiny dashboards for collaborator access.
- **Leadership & collaboration:** Mentored 10+ junior scientists; facilitated multi-institutional projects bridging wet-lab, computational, and clinical teams.

PROFESSIONAL EXPERIENCE

DOE, Joint Genome Institute, Research Scientist

(2014 – present)

- Led the design and analysis of large-scale genomics studies, utilizing isolate genomes, single-cell genomes and metagenomes, to characterize microbial diversity patterns and identify trends
- Provided scientific expertise in genomic surveillance methodologies, including developing and applying wet-lab and bioinformatic protocols aligning with organizational goals
- Managed interdisciplinary collaborations with academic researchers to implement genomic projects and effectively disseminate findings through reports and presentations
- Developed and maintained high-throughput bioinformatic pipelines for genomic data analysis, ensuring rapid and accurate responses to research queries
- Developed and maintained interactive web applications to disseminate research findings, enabling real-time data exploration for collaborators and stakeholders through publicly accessible institutional websites
- Mentored junior scientists in bioinformatic techniques, project design, and data interpretation, fostering knowledge transfer and capacity building

Virginia Polytechnic and State University, Postdoctoral Fellow

(2013 – 2014)

- Developed DNA-based detection tools to identify airborne viral pathogens (e.g., influenza) and validated assays for diagnostic applications
- Troubleshoot and optimized molecular workflows to enhance assay sensitivity and accuracy

University of Hawaii, Institute for Astrobiology, Postdoctoral Fellow

(2012 – 201

- Processed and analyzed microbial samples using Illumina sequencing to identify and characterize microbial diversity, applying rigorous QC procedures for data validation

University of Colorado, Ecology and Evolutionary Biology, Ph.D. Student

(2007 – 2012)

- Organized and led field campaigns for microbial bioaerosol sample collection, studying the spatiotemporal dynamics of airborne microorganisms
- Utilized qPCR to quantify microbial abundances, identifying potential health and environmental impacts of airborne microbes
- Prepared amplicon libraries for high-throughput sequencing platforms (Illumina and 454) to assess microbial community composition
- Provided bioinformatic and statistical analysis support for research projects beyond dissertation, contributing to cross-institutional projects

Advanced Bionutrition (Industry), Research Associate

(2005 – 2007)

- Designed and validated diagnostic assays (e.g., RT-PCR, ELISA) for detecting viral pathogens, supporting epidemiological monitoring in aquaculture
- Developed pathogen-specific vaccines for aquaculture species, addressing disease prevention and contributing to food security initiatives
- Maintained laboratory compliance with quality control standards and provided training to junior scientists

Aethlon Medical (Industry), Research Assistant

(2005 – 2005)

- Designed and optimized RT-qPCR assays for Hepatitis-C virus detection, enhancing diagnostic capabilities in clinical research

California State University, San Marcos, Research Assistant

(2003 – 2005)

- Developed and analyzed DNA microarrays to monitor gene expression in marine algae, advancing understanding of algal calcification processes
- Validated gene expression microarrays with RT-qPCR assays, ensuring data reliability and accuracy in environmental genomics research

EDUCATION

Ph.D., Ecology and Evolutionary Biology, University of Colorado (2012)

- Research Supervisor: Noah Fierer
- Dissertation: *The spatiotemporal dynamics of microbes in the near-surface atmosphere*, focusing on airborne microbial dispersal and environmental surveillance, with implications for public health and epidemiology

M.S., Biological Sciences, California State University, San Marcos (2006)

- Research Supervisor: Betsy Read
- Dissertation: *Identification of genes associated with calcification in a marine alga*, utilizing genomic techniques relevant to environmental microbiology and molecular surveillance

B.S., Biological Sciences, California State University, San Marcos (2003)

- Conducted undergraduate research in a molecular biology laboratory, developing foundational wet-lab bench skills and an introductory bioinformatics toolkit

TEACHING EXPERIENCE

2019 – DOE, Joint Genome Institute, Microbial Genomes and Metagenomes workshop

2013 – University of Hawaii, Hawaii Data Science Institute

2012 – University of Hawaii, Ali'i Teacher Workshop

2008 – University of Colorado, Teaching assistant for General Microbiology

2004 – California State University, San Marcos, Teaching assistant for Molecular Cell Biology

U.S. PATENTS

US Patent 8,221,767 "Antigenicity of infectious necrosis virus VP2 sub-viral particles expressed in yeast." Arun K. Dhar, **Robert M. Bowers** and F. C. Thomas Allnutt. Issued: July 17, 2012.

- Demonstrates contribution to vaccine development and pathogen diagnostics, aligning with disease monitoring and public health priorities

HONORS AND AWARDS

- NASA Astrobiology Fellowship, 2012 – 2013
- CIRES Graduate Student Research Fellowship, 2009 – 2013
- Alfred P. Sloan Foundation, Minority Ph.D. Program in Math, Science & Engineering, 2007 – 2011
- Graduate Research Initiative for Science Enhancement (RISE) Scholar, 2003 – 2005
- Undergraduate Research Initiative for Science Enhancement (RISE) Scholar, 2002 – 2003

INVITED PRESENTATIONS

- *Decoding Environmental Microbes: From Communities to Populations*. Estuary & Ocean Sciences (EOS) Center Seminar Series. California State University, San Francisco, San Francisco, CA. April 2025
- *Hot Spring genomics, one cell at a time*. Metagenome Forum. Universidad Miguel Hernandez, Alicante, Spain. March 2022
- *Microbial genomics, one cell at a time*. SACBS Seminar Series. University of Arizona, Tucson, AZ. March 2022
- *Microbial genomics at the JGI, an introduction*. Biological Sci Seminar Series. University of Texas El Paso, El Paso, TX. Nov 2021
- *Single-cell genomics at the JGI; from function-driven genomics to the analysis of populations*. Corteva Agriscience. Corteva, Indianapolis, IN. Aug 2019
- *Hot Spring genomics, one cell at a time*. University of Colorado, Boulder, CO. Jan 2019
- *Life at the boundary. Exploring the microbiology of the atmosphere*. University of Utah, Salt Lake City, UT. Mar 2017
- *Single-cell Genomics at the DOE Joint Genome Institute*. Labcyte Genomics Symposium, Edinburgh, Scotland. Sept 2016

SELECTED PUBLICATIONS

- **Bowers, R.M.**, Bennett, S., Riley, R., Villada, J., Ramalho Da Silva, I., Woyke, T., Frank, C. *Host species and geographic location shape microbial diversity and functional potential in the conifer needle microbiome*. (2025). *Microbiome*. Submitted
- **Bowers, R.M.***, Gonzalez-Pena, V.*, Wardhani K., Goudeau, D., Blow, M.J., Udway, D.K., Vill, A.C., Brito, I.L., Woyke, T., Malmstrom, R., Gawad, C. *scMicrobe PTA: Near Complete Genomes from Single Bacterial Cells*. (2024). *ISME Communications*. DOI: <https://doi.org/10.1093/ismeco/ycae085>. *Authors contributing equally
- **Bowers, R.M.**, *Gone with a trace: cataloguing the disappearing gut microbes*. (2023). *Nature Reviews Microbiology*. DOI: <https://doi.org/10.1038/s41579-023-00968-1>
- **Bowers, R.M.**, Nayfach, S., Schulz, F., Jungbluth, S.P., Ruhl, I., Lee, J., Goudeau, D., Elloe-Fadrosh, E.A., Stepanauskas, R., Dunfield, P., Malmstrom, R., Dunfield, P.F., Woyke, T., *Dissecting the dominant hot spring microbial populations based on community-wide sampling at single-cell resolution*. (2021). *The ISMEJ*. DOI: <https://doi.org/10.1038/s41396-021-01178-4>
- Grieb, A., **Bowers, R.M.**, Oggerin, M., Goudeau, D., Lee, J., Malmstrom, R.R., Woyke, T., Fuchs, B.M. *A pipeline for targeted metagenomics of environmental bacteria*. (2020). *Microbiome*. DOI: 10.1186/s40168-020-0790-7
- Doud, D.F.R.*, **Bowers, R.M.***, DeRaad, M.D., Deng, K., Tarver, A., Glasgow, E., Meulen, K.V., Fox, B., Deutsch, S., Yoshikuni, Y., Schulz, F., Northern, T., Hedlund, B.P., Singer, S.W., Ivannova, N., Woyke, T. *Identification of novel cellulose degrading bacteria using a function-driven single-cell genomics approach*. (2020). *The ISMEJ*. DOI: 10.1038/s41396-019-0557-y. *Authors contributing equally
- **Bowers, R.M.**, Doud, D., Woyke, T. *Analysis of single-cell genome sequences of bacteria and archaea*. (2017). *Emerging Topics in Life Sciences: Computational Biology*. DOI: 10.1042/ETLS20160028
- **Bowers, R.M.**, Kyrpides N., Stepanauskas R., Harmon-Smith M., et al.. *Minimum information about a single amplified genome (MISAG) and a metagenome-assembled genome (MIMAG) of bacteria and archaea*. (2017). *Nature Biotech*. DOI: 10.1038/nbt.3893
- Jungbluth, S.P., **Bowers, R.M.**, Lin, H-T., Cowen, J.P., Rappe, M.S. *A novel microbial assemblage inhabits fluids within mid-ocean ridge flank subsurface basalt*. (2016). *The ISME J*. DOI: 10.1038/ismej.2015.248

- **Bowers, R.M.**, Clum, A., Tice, H., Lim, J., Singh, K., Ciobanu, D., Yee Ngan, C., Cheng, J-F., Tringe, S.G., Woyke, T. *Impact of library preparation protocols and template quantity on the metagenomic reconstruction of a mock microbial community.* (2015). *BMC Genomics*. DOI: 10.1186/s12864-015-2063-6
- **Bowers, R.M.**, Hamady, M., Knight, R., Lauber, C., Wiedinmyer, C., Fall, R., Haller, A.G., Fierer, N. *Characterization of airborne microbial communities at a high-elevation site and their potential to act as atmospheric ice nuclei.* (2009). *Applied and Environmental Microbiology*. DOI: 10.1128/AEM.00447-09
- Wiedinmyer, C., **Bowers, R.**, Fierer, N., Horanyi, E., Hannigan, M., Hallar, A.G., McCubbin, I.B., Baustian, K. *The contribution of biological particles to observed particulate organic carbon at a remote high altitude site.* (2009). *Atmospheric Environment*. DOI: 10.1016/j.atmosenv.2009.06.012
- Dhar, A. K., **Bowers, R. M.**, Rowe, C.G., Allnutt, F. C. T. *Expression of human oncogene c-myc epitope on infectious pancreatic necrosis virus (IPNV) VP2 capsid protein sub-viral particle (sVLP) and the immunogenicity of the chimeric sVLP in rainbow trout.* (2010). *Vaccine*. DOI: 10.1016/j.antiviral.2009.12.009
- Dhar, A.K., **Bowers, R.M.**, Licon, K.S., Veazey, G., Read, B. *Validation of reference genes for quantitative measurement of immune gene expression in shrimp.* (2009). *Journal of Molecular Immunology*. DOI: 10.1016/j.molimm.2009.02.020
- Dhar, A.K., **Bowers, R.M.**, Licon, K.S., LaPatra, S.E. *Detection and quantification of infectious hematopoietic necrosis virus in rainbow trout (*Oncorhynchus mykiss*) by SYBR Green real-time reverse transcriptase-polymerase chain reaction.* (2008). *Journal of Virological Methods*. DOI: 10.1016/j.jviromet.2007.08.026
- **Bowers, R.M.**, LaPatra, S.E., Dhar, A.K. *Detection and quantification of Infectious Pancreatic Necrosis Virus by real-time reverse transcriptase-polymerase chain reaction using lethal and non-lethal tissue sampling.* (2008). *Journal of Virological methods*. DOI: 10.1016/j.jviromet.2007.09.003
- Allnutt, F.C.T., **Bowers, R.M.**, Rowe, C.G., Vakharia, V.N., LaPatra, S.E., Dhar, A.K. *Antigenicity of infectious pancreatic necrosis virus capsid protein VP2 virus-like particles expressed in yeast.* (2007). *Antiviral Research*. DOI: 10.1016/j.vaccine.2007.04.068
- Quinn, P., **Bowers, R.M.**, Zhang, X., Wahlund, T.M., Fanelli, M.F., Olszova, D., Read, B.A. *cDNA Microarrays as a Tool for Identification of Biomineralization Proteins in the Coccolithophorid *Emiliana huxleyi* (Haptophyta).* (2006). *Applied and Environmental Microbiology*. 10.1128/AEM.00343-06.
- Nguyen, B., **Bowers, R.M.**, Wahlund, T.M., Read, B.A. *Suppressive Subtractive Hybridization and Differences in Gene Content Between Calcifying and Non-Calcifying Cultures of *Emiliana huxleyi* Strain 1516.* (2005). *Applied and Environmental Microbiology*. DOI: 10.1128/AEM.71.5.2564-2575.2005